Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/15/2005
PI Name:	Eiceman, Gary A. Ph.D.		
Project Title:	Micro-Machined Drift Tubes for Ion Mobility Spectrometry: Studies Toward a Microbial Analyzer for Manned Spacecraft		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced environmental monitoring and control		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	88003	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-01
Start Date:	01/01/2003	**End Date:**	12/31/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Smith, Geoff B (Dept of Biology)		
Grant/Contract No.:	NAG9-1477		
Performance Goal No.:			
Performance Goal Text:			
	The goal of this research program is to provide capabilities for the detection and identification of microorganisms on manned spacecraft through chemotaxonomy with chemical instrumentation. Essential engineering considerations include small size, low weight, and low power. The requirements for performance include automated operation, response toward microorganisms relevant to spacecraft, and suitability for sampling air, water, and certain materials. The chemical instrument is based on pyrolysis of microorganisms with vapor characterization by gas chromatography/mobility spectrometry and technology upgrades will be made with micro-machined mobility spectrometry and small gas chromatographs. These new micro-scale drift tubes have been operated under high field radio frequency (RF) regimes without ion shutters and aperture grids yielding low detection limits and high mechanical integrity. A research program to meet the goal of small size with high bio-analytical performance is planned with four stages of effort including:		

Task Description:	refinement of existing instrumentation, creation of a library from the determination of microorganisms (bacteria, viruses, fungi) by pyrolysis-gas chromatography/micro-machined mobility spectrometry, preparation of methodology for data processing, and creation of procedures for automated interpretation of data. These advances will be made using drift tubes barely the size of a quarter coin to detect the volatile organic compounds that are formed as thermal decomposition products from pyrolysis of microorganisms. The drift tube will be joined to newly available, low mass, temperature programmable gas chromatographs and small Curie Point pyrolysis units. The combined pyrolysis-gas chromatograph/micro-mobility spectrometer and resultant experiments will permit a clear assessment of this technology for potential service on spacecraft. All studies will be completed in the laboratory where experience and technology of micro-drift tubes have been matured and where all aspects of this project (separations, mobility spectrometry, bacterial measurements by mass spectrometry as a validation technique, and chemometric analysis of data) exist.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Between 1979 and 2000, more than 10 million cases of sepsis were reported in the United States having steadily risen from about 164,000 to nearly 660,000 annual cases within the past two decades. Even at the end of the 20th century, the in-hospital mortality rate due to sepsis was still as high as 17.9 % so that according to the National Center for Health Statistics, sepsis is the 10th leading cause of death overall in the United States. As a major public health problem, sepsis consumes more than $17 billion in healthcare resources annually in the United States. Gram-positive bacteria and fungal organisms were identified as increasingly common causes of sepsis. [1] This example shows that besides in space, a fast detection of microorganisms is also urgently needed on earth. The technology and methods developed for the detection of microorganisms on manned space crafts show advantages from which not only medical applications could benefit, but which are also needed on earth for e.g. environmental and even homeland security uses: i.e. short total analysis times, low cost, convenience, low power, low weight and good analytical performance. Thus, the advances made in this project will have direct impact on the rapid determination of bacteria, viruses and fungi. [1] Martin GS, Mannino DM, Eaton S, Moss. M. The epidemiology of sepsis in the United States from 1979 through 2000. N Engl J Med 2003, 348:1546-1554
Task Progress:	Micro-fabricated drift tubes for differential mobility spectrometry (DMS) were used as detectors for gas chromatography (GC) separations of vapors from pyrolysis (Py) of bacteria and provided response through gas phase ion chemistry at ambient pressure. Topographical plots of ion intensity, compensation voltage, and retention time were obtained simultaneously for positive and negative ions from py-GC-DMS analysis of bacteria. Thus, each Py-GC-DMS experiment included selectivity from chemical reactions in air through reagent ions H+(H2O)n and O2-(H2O)n. Chemical information was available throughout the span of elution temperatures from 35 to 200 °C extended here over previous designs with improved temperature control of the DMS analyzer. Nine bacteria, amongst which were representatives of Gram-positive (e.g., M. luteus), Gram-positive spore formers (e.g., B. subtilis) and Gram-negative (e.g., K. pneumoniae) species, were analyzed by Py-GC/DMS and plots of intensity versus retention time were extracted from topographic plots at six different compensation voltages. These chromatograms contained chemical information from differential mobility analysis of ions in the elution profile and their distinctive patterns were used with backpropagation neural networks to categorize bacteria. Adjusted values (AV) for identification of bacteria were dependent on the compensation voltage used and average AV were up to 75 % for vegetative cells and up to 78 % for spores. Assignments of peak identity in the Py-GC-DMS plots were supported by parallel experiments using Py-GC with electron impact mass selective detection.
Bibliography Type:	Description: (Last Updated: 06/15/2005)
Articles in Peer-reviewed Journals	Schmidt, H.; Tadjimukhamedov, F.; Mohrenz, I.V.; Smith, G.B.; and Eiceman, G.A. "Microfabricated Differential Mobility Spectrometry with Pyrolysis Gas Chromatography for Chemical Characterization of Bacteria" Anal. Chem. 2004, 76, 5208-5217 , Sep-2004
Papers from Meeting Proceedings	Schmidt, H.; Tadjimukhamedov, F., Mohrenz, I.V., Smith, G.B.; and Eiceman, G.A. "Characterization of Bacteria Using Micro-Fabricated Differential Mobility Spectrometry with Pyrolysis Gas Chromatography" 13th International Conference on Ion Mobility Spectrometry, July 25-29, 2004, Gatlinburg, TN, USA Jul-2004
Papers from Meeting Proceedings	Wang, M.; Prasad, S.; Schmidt, H.; Douglas, K.; Snyder, A.P.; Smith, G.B.; and Eiceman, G.A. "Ion Mobility Spectrometry with Pyrolysis Gas Chromatography for Chemical Characterization of Bacteria" 13th International Conference on Ion Mobility Spectrometry, July 25-29, 2004, Gatlinburg, TN, USA Jul-2004
Presentation	Bonano, E.E.; Carothers, M.G.C.; Schmidt, H.; Prasad, S.; Wang, M.; Douglas, K.; Snyder, A.P.; Smith, G.B.; Eiceman, G.A "Pyrolysis Gas Chromatography Ion Mobility Spectrometry: A powerful tool for a fast chemical characterization of bacteria" SACNAS National Conference, Austin, TX, USA, October 21 - 24, 2004 Oct-2004
Presentation	Eiceman, G.A.; and Schmidt, H. "Characterization of Airborne Substances or Micro-organisms using technology involving differential mobility spectrometry" Strategies for the Detection of Low Vapor Pressure Chemicals, Department of Homeland Security, Arlington, VA, USA, April 6 - 7, 2004 Apr-2004
Presentation	Schmidt, H.; Tadjimukhamedov, F.; Smith, G.B.; Mohrenz, I.; Eiceman, G.A. "Pyrolysis Gas Chromatography Differential Mobility Spectrometry of Bacteria" Habitat 2004, Orlando, FL, USA, January 4 - 7, 2004 Jan-2004
Presentation	Schmidt, H.; Tadjimukhamedov, F.; Prasad, S.; Smith, G.B.; and Eiceman, G.A. "Pyrolysis Gas Chromatography with Micro-Fabricated Differential Mobility Spectrometry: A Powerful Combination for a Fast and Sensitive Characterization of Bacteria" Pittcon 2005, Orlando, FL, Feb 27 - Mar 04, 2005 Feb-2005

Presentation	Prasad, S.; Schmidt, H.; Wang, M.; Douglas, K.; Snyder, A.P.; Smith, G.B.; and Eiceman, G.A. "Biomarkers for characterization of gram negative and gram positive bacteria by coupling pyrolysis gas chromatography to ion mobility spectrometry" Pittcon 2005, Orlando, FL, Feb 27 - Mar 04, 2005 Feb-2005